METRIC

MIL-STD-2411-1
w/CHANGE 4
20 August 2020
SUPERSEDING
MIL-STD-2411-1
w/CHANGE 3
1 December 2009

DEPARTMENT OF DEFENSE INTERFACE STANDARD

REGISTERED DATA VALUES FOR RASTER PRODUCT FORMAT

AMSC N/A **AREA GINT**

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FOREWORD

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Comments, suggestions or questions should be addressed to the National Geospatial-Intelligence Agency (NGA) National Center for Geospatial Intelligence Standards (NCGIS), Mail Stop N32, 7500 GEOINT Drive, Springfield, VA 22150, or emailed to ncgis-mail@nga.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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SUMMARY OF CHANGE 4 MODIFICATIONS

The following modifications to MIL-STD-2411-1 Change Notice 3, 1 December 2009, have been made:

Paragraph	Modification
2.1	Added paragraph
5.1.4	Changed table (codes added)
5.1.5	Changed text
5.1.7	Changed text, removed table
5.2.1	Changed table (codes added)
5.3.2.1	Changed table (Descriptions for Attribute 21, Parameter 1 and Attribute 24, Parameters 3 and 7)
5.4	URL changed

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1 SCOPE

1.1 Scope

This document is a companion document to the military standard for Raster Product Format (RPF), MIL-STD-2411. It contains all registered data values referenced in the standard.

1.2 Purpose

This document is intended to supply producers and users of RPF products with the registered data values necessary to produce and interpret RPF data.

1.3 Application

The Military Departments, Office of the Secretary of Defense, Organizations of the Joint Chiefs of Staff, and the Defense Agencies of the Department of Defense (collectively known as DoD components) use the information in this document in preparing and accessing digital geographic data required or specified to be in RPF.

1.4 Security

This standard is UNCLASSIFIED. The registered data values presented herein may be used for classified processing where appropriate security provisions are added.

2 APPLICABLE DOCUMENTS

2.1 General

The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2411 Military Standard for Raster Product Format, with Change Notice 2

(Copies of federal and military specifications, standards, and handbooks are available at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications

The following other government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NGA.STND.0036_1.0.0_WGS84, Department of Defense World Geodetic System 1984 : Its Definition and Relationships with Local Geodetic Systems, Version 1.0.0, 8 July 2014.

Copies of this standard are available at https://nsgreg.nga.mil.

2.3 Non-Government publications

This section is not applicable to this standard.

2.4 Order of precedence

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. However, nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 DEFINITIONS

3.1 Acronyms and abbreviations

a. ASCII American Standard Code for Information Interchange
b. CMYK Cyan, Magenta, Yellow and Black
c. NGA National Geospatial-Intelligence Agency
d. NITF National Imagery Transmission Format
e. RGB Red, Green, Blue
f. RGBM Red, Green, Blue and Monochromatic
g. RPF Raster Product Format
h. VQ Vector Quantization

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4 GENERAL REQUIREMENTS

This document provides all registered data values to be used in products that conform to the RPF military standard (MIL-STD-2411). The data includes identification codes, attribute and parameter tables, and other tables required for RPF products.

RPF includes several distinct file types (e.g., [table of contents file], [frame file]). Some of the registered data values found in this document are unique to particular file types, and others are found in several different file types. The specific registered values are listed in section 5, Detailed Requirements. The format is such that new registered values can easily be added to specific subsections. Where applicable, unused identification codes have been marked as reserved for future use.

Note: Change 4 of MIL-STD-2411-1 resulted in the addition of 18 data series codes to the table in 5.1.4.

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5 DETAILED REQUIREMENTS

Section 5 is a listing of the registered data values that are used in RPF products.

Each of the subsections in section 5 will include a description of the registered data unique to that subsection, the code or ID corresponding to the data field, and, in some cases the size and data type of that code. The data types used are listed in TABLE I and defined in MIL-STD-2411. All registered data field types and sizes are defined in MIL-STD-2411.

TABLE I. Data types and their abbreviations

Data Type	**Abbreviation**
ASCII string	asci
Bit string	bits
Byte string	byte
Boolean	bool
Integer (signed)	ints
Integer (unsigned)	uint
Real	real

Section 5.1 describes global data. Global data is defined here as data that can occur in more than one type of RPF file (i.e., <section/component id> values are necessary in all RPF files). Section 5.2 describes data that is used only in the [table of contents file]. Section 5.3 describes data that is used only in the [frame file].

5.1 Global data

Data in this section will pertain to information that may be used in more than one type of RPF file.

5.1.1 Section/Component ID codes

The following are the valid <section/component id> codes currently defined in MIL-STD-2411.

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TABLE II. Section/component codes

<section/component id>	[section/component]
1-127	Reserved for future standardization
128	header component
129	location component
130	coverage section subheader
131	compression section subheader
132	compression lookup subsection
133	compression parameter subsection
134	color/grayscale section subheader
135	colormap subsection
136	image description subheader
137	image display parameters subheader
138	mask subsection
139	color converter subsection
140	spatial data subsection
141	attribute section subheader
142	attribute subsection
143	explicit areal coverage table
144	related images section subheader
145	related images subsection
146	replace/update section subheader
147	replace/update table
148	boundary rectangle section subheader
149	boundary rectangle table
150	frame file index section subheader
151	frame file index subsection
152	color table index section subheader
153	color table index record
154-65535	Reserved for future standardization

5.1.2 Color/Grayscale section codes

The following sections define the valid [color/grayscale section] codes currently defined for RPF products, which include <color/grayscale table id> codes and <color converter> codes.

The two types of table id's can be defined as traditional color/grayscale tables that contain values for the various color planes and/or bands within an image, and color converter tables, which define a mapping between two color/grayscale tables. Sections 5.1.2.1 and 5.1.2.2 describe the two [color/grayscale] section codes in more detail.

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5.1.2.1 Color table codes

Each of the color/grayscale tables identified in the table below represents a particular organization of the color planes in the table. For example, a color table containing Red, Green and Blue (RGB) color planes would have an id of one indicating an RGB arrangement of the table. Examples of other arrangements of the color table that are currently registered in RPF include Red, Green, Blue, Monochromatic (RGBM); Monochromatic (M); and Cyan Magenta Yellow, Black (CMYK), which is often used in color printing.

TABLE III. Color table codes

<color/grayscale table id>	**Description**
1 2 3 4	RGB RGBM M CMYK
5-65535	Reserved for future standardization

5.1.2.2 Color conversion codes.

The color conversion table maps the colors from one color/grayscale table into another color/grayscale table. If a user is unable, or does not choose, to use the primary color/grayscale table in an RPF file, the color conversion table can be used to map the primary colors into the closest matching colors within the alternate table. These names used to describe the <color/grayscale table id>'s begin with the letters CCT/. The number immediately after the slash (/) is the <color/grayscale table id> of the primary color table, and the number after the arrow (->) is the alternate color table id. In cases where the primary and alternate color tables contain the same color planes, the numbers after the slashes will be the same. In these cases the color converter table is often used to map a larger number of colors in the primary color table to a smaller number of colors in the alternate color table.

As an example of the use of a color conversion table, assume that an RPF file has two <color/grayscale table id> values and one <color converter table id> value. The two [color/grayscale table] are RGBM (<color/grayscale table id> = 2) and therefore the appropriate [color converter table would map the primary RGBM into

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the alternate RGBM, (<color converter table id> = 5). The color conversion table is a mapping of the colors from the primary RGBM color table to the alternate RGBM color table as shown in FIGURE 1. A call to display a color X from the primary color table will be routed to the color conversion table, where the index of X will be mapped to the index of Y, a color in the alternate color table.

TABLE IV. Color conversion codes

<color converter table id>	**Description**
1	CCT/1->1
2	CCT/1->2
3	CCT/1->3
4	CCT/1->4
5	CCT/2->2
6	CCT/2->3
7	CCT/2->4
8	CCT/3->3
10-65535	Reserved for future standardization

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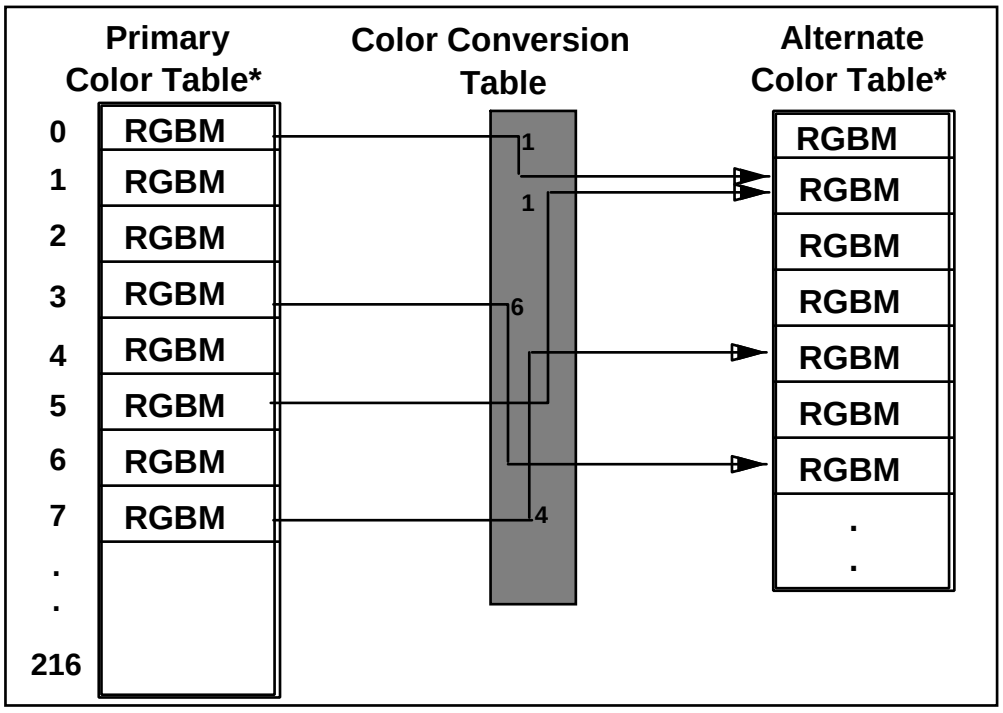


FIGURE 1. Color mapping between primary and alternate color tables

5.1.3 Zone codes

Zone codes are used in conjunction with the data series codes (5.1.4) to construct the <data series and zone> codes in 5.1.5. For products where zone code and/or data series is not applicable, the <data series and zone code> shall consist of a 3-byte descriptor applicable to a particular product. No <data series and zone> codes are currently registered for such products.

TABLE V. Zone codes

Zone Number	Equatorward Latitude	Midpoint Latitude	Poleward Latitude
1,A	0°	22.94791772°	32°
2,B	32°	41.12682127°	48°
3,C	48°	52.28859923°	56°
4,D	56°	60.32378942°	64°
5,E	64°	66.09421768°	68°
6,F	68°	70.10896259°	72°
7,G	72°	74.13230145°	76°
8,H	76°	78.17283750°	80°
9,J	80°	———	90°

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5.1.3.1 Zone codes for DTED data

Zone codes are used in conjunction with the data series codes (5.1.4) to construct the <data series and zone> codes in 5.1.5. The zone codes for RPF data that is zoned similar to DTED data have the following zone codes.

TABLE VI. Zone codes for DTED data

Zone Number	Equatorward Latitude	Poleward Latitude
1,6	0°	50°
2,7	50°	70°
3,8	70°	75°
4,9	75°	80°
5,A	80°	90°

5.1.4 Data series codes

The data series codes are used in conjunction with the zone codes (5.1.3) to construct the <data series and zone> codes in 5.1.5. The specific methodology for creating the data series and zone codes is described in MIL-STD-2411.

TABLE VII. Data series codes

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
A1	CM	1:10,000	Combat Charts/LPCs, 1:10,000 scale	CADRG
A2	CM	1:25,000	Combat Charts/LPCs, 1:25,000 scale	CADRG
A3	CM	1:50,000	Combat Charts/LPCs, 1:50,000 scale	CADRG
A4	CM	1:100,000	Combat Charts/LPCs, 1:100,000 scale	CADRG
A5	CM	1:12,500	Combat Charts/LPCs, 1:12,500 scale	CADRG
A6	CM	Various	Combat Charts/LPCs, various scales	CADRG
AT	ATC	1:200,000	Series 200 Air Target Chart	CADRG
C1	CG	1:10,000	City Graphics	CADRG
C2	CG	1:10,560	City Graphics	CADRG
C3	CG	1:11,000	City Graphics	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
C4	CG	1:11,800	City Graphics	CADRG
C5	CG	1:12,000	City Graphics	CADRG
C6	CG	1:12,500	City Graphics	CADRG
C7	CG	1:12,800	City Graphics	CADRG
C8	CG	1:14,000	City Graphics	CADRG
C9	CG	1:14,700	City Graphics	CADRG
CA	CG	1:15,000	City Graphics	CADRG
CB	CG	1:15,500	City Graphics	CADRG
CC	CG	1:16,000	City Graphics	CADRG
CD	CG	1:16,666	City Graphics	CADRG
CE	CG	1:17,000	City Graphics	CADRG
CF	CG	1:17,500	City Graphics	CADRG
CG	CG	Various	City Graphics	CADRG
CH	CG	1:18,000	City Graphics	CADRG
CJ	CG	1:20,000	City Graphics	CADRG
CK	CG	1:21,000	City Graphics	CADRG
CL	CG	1:21,120	City Graphics	CADRG
CM	CM	Various	Combat Charts	CADRG
CN	CG	1:22,000	City Graphics	CADRG
CO	CO	Various	Coastal Charts	CADRG
CP	CG	1:23,000	City Graphics	CADRG
CQ	CG	1:25,000	City Graphics	CADRG
CR	CG	1:26,000	City Graphics	CADRG
CS	CG	1:35,000	City Graphics	CADRG
CT	CG	1:36,000	City Graphics	CADRG
D1[1]	-	100m	Elevation Data from DTED Level 1	CDTED
D2[1]	-	30m	Elevation Data from DTED Level 2	CDTED
EG	NARC	1:11,000,000	North Atlantic Route Chart	CADRG
ES	SEC	1:500,000	VFR Sectional	CADRG
ET	SEC	1:250,000	VFR Sectional Insets	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
F1	TFC-1	1:250,000	Transit Flying Chart (TBD #1)	CADRG
F2	TFC-2	1:250,000	Transit Flying Chart (TBD #2)	CADRG
F3	TFC-3	1:250,000	Transit Flying Chart (TBD #3)	CADRG
F4	TFC-4	1:250,000	Transit Flying Chart (TBD #4)	CADRG
F5	TFC-5	1:250,000	Transit Flying Chart (TBD #5)	CADRG
GN	GNC	1:5,000,000	Global Navigation Chart	CADRG
HA	HA	Various	Harbor and Approach Charts	CADRG
I1	-	10m	Imagery, 10 meter resolution	CIB
I2	-	5m	Imagery, 5 meter resolution	CIB
I3	-	2m	Imagery, 2 meter resolution	CIB
I4	-	1m	Imagery, 1 meter resolution	CIB
I5	-	.5m	Imagery, .5 (half) meter resolution	CIB
IV	-	Various > 10m	Imagery, > 10 meter resolution	CIB
JA	JOG-A	1:250,000	Joint Operations Graphic - Air	CADRG
JG	JOG	1:250,000	Joint Operations Graphic	CADRG
JN	JNC	1:2,000,000	Jet Navigation Chart	CADRG
JO	OPG	1:250,000	Operational Planning Graphic	CADRG
JR	JOG-R	1:250,000	Joint Operations Graphic - Radar	CADRG
K1	ICM	1:8,000	Image City Maps	CADRG
K2	ICM	1:10,000	Image City Maps	CADRG
K3	ICM	1:10,560	Image City Maps	CADRG
K7	ICM	1:12,500	Image City Maps	CADRG
K8	ICM	1:12,800	Image City Maps	CADRG
KB	ICM	1:15,000	Image City Maps	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
KE	ICM	1:16,666	Image City Maps	CADRG
KM	ICM	1:21,120	Image City Maps	CADRG
KR	ICM	1:25,000	Image City Maps	CADRG
KS	ICM	1:26,000	Image City Maps	CADRG
KU	ICM	1:36,000	Image City Maps	CADRG
L1	LFC-1	1:500,000	Low Flying Chart (TBD #1)	CADRG
L2	LFC-2	1:500,000	Low Flying Chart (TBD #2)	CADRG
L3	LFC-3	1:500,000	Low Flying Chart (TBD #3)	CADRG
L4	LFC-4	1:500,000	Low Flying Chart (TBD #4)	CADRG
L5	LFC-5	1:500,000	Low Flying Chart (TBD #5)	CADRG
LF	LFC-FR (Day)	1:500,000	Low Flying Chart (Day)- Host Nation	CADRG
LN	LFC (Night)	1:500,000	Low Flying Chart (Night) - Host Nation	CADRG
M1	MIM	Various	Military Installation Map (1:100K & larger)	CADRG
M2	MIM	Various	Military Installation Map (smaller than 1:100K)	CADRG
MH	MIM	1:25,000	Military Installation Maps	CADRG
MI	MIM	1:50,000	Military Installation Maps	CADRG
MJ	MIM	1:100,000	Military Installation Maps	CADRG
MM	-	Various	(Miscellaneous Maps & Charts)	CADRG
OA	OPAREA	Various	Naval Range Operating Area Chart	CADRG
OH	VHRC	1:1,000,000	VFR Helicopter Route Chart	CADRG
ON	ONC	1:1,000,000	Operational Navigation Chart	CADRG
OW	WAC	1:1,000,000	High Flying Chart – Host Nation	CADRG
P1[2]	-	1:25,000	Special Military Map - Overlay	CADRG
P2	-	1:25,000	Special Military Purpose	CADRG
P3	-	1:25,000	Special Military Purpose	CADRG
P4	-	1:25,000	Special Military Purpose	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
P5[2]	-	1:50,000	Special Military Map - Overlay	CADRG
P6	-	1:50,000	Special Military Purpose	CADRG
P7	-	1:50,000	Special Military Purpose	CADRG
P8	-	1:50,000	Special Military Purpose	CADRG
P9[2]	-	1:100,000	Special Military Map - Overlay	CADRG
PA	-	1:100,000	Special Military Purpose	CADRG
PB	-	1:100,000	Special Military Purpose	CADRG
PC	-	1:100,000	Special Military Purpose	CADRG
PD[2]	-	1:250,000	Special Military Map - Overlay	CADRG
PE	-	1:250,000	Special Military Purpose	CADRG
PF	-	1:250,000	Special Military Purpose	CADRG
PG	-	1:250,000	Special Military Purpose	CADRG
PH	-	1:500,000	Special Military Map - Overlay	CADRG
PI	-	1:500,000	Special Military Purpose	CADRG
PJ	-	1:500,000	Special Military Purpose	CADRG
PK	-	1:500,000	Special Military Purpose	CADRG
PL	-	1:1,000,000	Special Military Map - Overlay	CADRG
PM	-	1:1,000,000	Special Military Purpose	CADRG
PN	-	1:1,000,000	Special Military Purpose	CADRG
PO	-	1:1,000,000	Special Military Purpose	CADRG
PP	-	1:2,000,000	Special Military Map - Overlay	CADRG
PQ	-	1:2,000,000	Special Military Purpose	CADRG
PR	-	1:2,000,000	Special Military Purpose	CADRG
PS	-	1:5,000,000	Special Military Map - Overlay	CADRG
PT	-	1:5,000,000	Special Military Purpose	CADRG
PU	-	1:5,000,000	Special Military Purpose	CADRG
PV	-	1:5,000,000	Special Military Purpose	CADRG
R1	-	1:50,000	Range Charts	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
R2	-	1:100,000	Range Charts	CADRG
R3	-	1:250,000	Range Charts	CADRG
R4	-	1:500,000	Range Charts	CADRG
R5	-	1:1,000,000	Range Charts	CADRG
RC	RGS-100	1:100,000	Russian General Staff Maps	CADRG
RL	RGS-50	1:50,000	Russian General Staff Maps	CADRG
RR	RGS-200	1:200,000	Russian General Staff Maps	CADRG
RV	Riverine	1:50,000	Riverine Map 1:50,000 scale	CADRG
RW	Riverine	1:100,000	Riverine Map 1:100,000 scale	CADRG
SC	TLM100S	1:100,000	Base 100K TLM with overlay	CADRG
SL	TLM50S	1:50,000	Base 50K TLM with overlay	CADRG
SQ	TLM24S	1:24,000	Base 24K TLM with overlay	CADRG
SR	TLM200S	1:200,000	Base 200K TLM with overlay	CADRG
SS	TLM250S	1:250,000	Base 250K TLM with overlay	CADRG
ST	TLM25S	1:25,000	Base 25K TLM with overlay	CADRG
SU	TLM10S	1:10,000	Base 10K TLM with overlay	CADRG
TC	TLM100	1:100,000	Base TLM 1:100,0000 scale	CADRG
TF	TFC (Day)	1:250,000	Transit Flying Chart (Day)	CADRG
TL	TLM50	1:50,000	Base TLM 1:50,000 scale	CADRG
TN	TFC (Night)	1:250,000	Transit Flying Chart (Night) - Host nation	CADRG
TP	TPC	1:500,000	Tactical Pilotage Chart	CADRG
TQ	TLM24	1:24,000	Base TLM 1:24,000 scale	CADRG
TR	TLM200	1:200,000	Base TLM 1:200,000 scale	CADRG
TS	TLM250	1:250,000	Base TLM 1:250,000 scale	CADRG
TT	TLM25	1:25,000	Base TLM 1:25,000 scale	CADRG
TU	TLM10	1:10,000	Base TLM 1:10,000 scale	CADRG
UC	ALT100	1:100,000	Alternate TLM 100,000 scale	CADRG
UL	ALT50	1:50,000	Alternate TLM 1:50,000	CADRG
UR	ALT200	1:200,000	Alternate TLM 1:200,000	CADRG
US	ALT250	1:250,000	Alternate TLM 1:250,000	CADRG
UT	ALT25	1:25,000	Alternate TLM 1:25,000	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
UU	ALT10	1:10,000	Alternate TLM 1:10,000	CADRG
UV	ALTV	Various	Alternate TLM various scale	CADRG
V1	HRC Inset	1:50,000	Helicopter Route Chart Inset	CADRG
V2	HRC Inset	1:62,500	Helicopter Route Chart Inset	CADRG
V3	HRC Inset	1:90,000	Helicopter Route Chart Inset	CADRG
V4	HRC Inset	1:250,000	Helicopter Route Chart Inset	CADRG
VH	HRC	1:125,000	Helicopter Route Chart	CADRG
VN	VNC	1:500,000	Visual Navigation Charts	CADRG
VT	VTAC	1:250,000	VFR Terminal Area Chart	CADRG
VU	VTAC	1:1,000,000	VFR Terminal Area Chart	CADRG
WA	-	1:250,000	IFR Enroute Low	CADRG
WB	-	1:500,000	IFR Enroute Low	CADRG
WC	-	1:750,000	IFR Enroute Low	CADRG
WD	-	1:1,000,000	IFR Enroute Low	CADRG
WE	-	1:1,500,000	IFR Enroute Low	CADRG
WF	-	1:2,000,000	IFR Enroute Low	CADRG
WG	-	1:2,500,000	IFR Enroute Low	CADRG
WH	-	1:3,000,000	IFR Enroute Low	CADRG
WI	-	1:3,500,000	IFR Enroute Low	CADRG
WK	-	1:4,500,000	IFR Enroute Low	CADRG
XD	-	1:1,000,000	IFR Enroute High	CADRG
XE	-	1:1,500,000	IFR Enroute High	CADRG
XF	-	1:2,000,000	IFR Enroute High	CADRG
XG	-	1:2,500,000	IFR Enroute High	CADRG
XH	-	1:3,000,000	IFR Enroute High	CADRG
XI	-	1:3,500,000	IFR Enroute High	CADRG
XJ	-	1:4,000,000	IFR Enroute High	CADRG
XK	-	1:4,500,000	IFR Enroute High	CADRG
Y9	-	1:16,500,000	IFR Enroute Area	CADRG
YA	-	1:250,000	IFR Enroute Area	CADRG

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TABLE VII. Data series codes - Continued

Code	Series Abbr.	Scale/resolution	Data series	RPF data type
YB	-	1:500,000	IFR Enroute Area	CADRG
YC	-	1:750,000	IFR Enroute Area	CADRG
YD	-	1:1,000,000	IFR Enroute Area	CADRG
YE	-	1:1,500,000	IFR Enroute Area	CADRG
YF	-	1:2,000,000	IFR Enroute Area	CADRG
YI	-	1:3,500,000	IFR Enroute Area	CADRG
YJ	-	1:4,000,000	IFR Enroute Area	CADRG
YZ	-	1:12,000,000	IFR Enroute Area	CADRG
Z8	-	1:16,000,000	IFR Enroute High/Low	CADRG
ZA	-	1:250,000	IFR Enroute High/Low	CADRG
ZB	-	1:500,000	IFR Enroute High/Low	CADRG
ZC	-	1:750,000	IFR Enroute High/Low	CADRG
ZD	-	1:1,000,000	IFR Enroute High/Low	CADRG
ZE	-	1:1,500,000	IFR Enroute High/Low	CADRG
ZF	-	1:2,000,000	IFR Enroute High/Low	CADRG
ZG	-	1:2,500,000	IFR Enroute High/Low	CADRG
ZH	-	1:3,000,000	IFR Enroute High/Low	CADRG
ZI	-	1:3,500,000	IFR Enroute High/Low	CADRG
ZJ	-	1:4,000,000	IFR Enroute High/Low	CADRG
ZK	-	1:4,500,000	IFR Enroute High/Low	CADRG
ZT	-	1:9,000,000	IFR Enroute High/Low	CADRG
ZV	-	1:10,000,000	IFR Enroute High/Low	CADRG
ZZ	-	1:12,000,000	IFR Enroute High/Low	CADRG

[1] For user-specified applications of RPF that have not been approved for joint use

[2] Use is restricted to Military Services only, for operational overlay graphics

5.1.5 Data series and zone codes

The <data series and zone> codes are three digit alphanumeric codes derived from the two digit Data Series Code (see 5.1.4) and the one-digit Zone Code (see 5.1.3). The following are examples of valid codes currently defined for RPF data. Other valid <data series and zone> codes (for city graphics, etc.) can be created by

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combining data series codes with the ARC zonecodes in the same fashion.

TABLE VIII. Northern hemisphere – data series/zone codes

Data series	ZONE								
	1	2	3	4	5	6	7	8	9
GN	GN1	GN2	GN3	GN4	N/A	N/A	GN7	N/A	GN9
JN	JN1	JN2	JN3	JN4	JN5	JN6	JN7	JN8	JN9
ON	ON1	ON2	ON3	ON4	ON5	ON6	ON7	ON8	ON9
TP	TP1	TP2	TP3	TP4	TP5	TP6	TP7	TP8	TP9
LF	LF1	LF2	LF3	LF4	LF5	LF6	LF7	LF8	LF9
JG	JG1	JG2	JG3	JG4	JG5	JG6	JG7	JG8	JG9
JA	JA1	JA2	JA3	JA4	JA5	JA6	JA7	JA8	JA9
JR	JR1	JR2	JR3	JR4	JR5	JR6	JR7	JR8	JR9
TF	TF1	TF2	TF3	TF4	TF5	TF6	TF7	TF8	TF9
AT	AT1	AT2	AT3	AT4	AT5	AT6	AT7	AT8	AT9
TC	TC1	TC2	TC3	TC4	TC5	TC6	TC7	TC8	TC9
TL	TL1	TL2	TL3	TL4	TL5	TL6	TL7	TL8	TL9
TT	TT1	TT2	TT3	TT4	TT5	TT6	TT7	TT8	TT9
TQ	TQ1	TQ2	TQ3	TQ4	TQ5	TQ6	TQ7	TQ8	TQ9
CG	CG1	CG2	CG3	CG4	CG5	CG6	CG7	CG8	CG9
CM	CM1	CM2	CM3	CM4	CM5	CM6	CM7	CM8	CM9
A1	A11	A12	A13	A14	A15	A16	A17	A18	A19
A2	A21	A22	A23	A24	A25	A26	A27	A28	A29
A3	A31	A32	A33	A34	A35	A36	A37	A38	A39
MI	MI1	MI2	MI3	MI4	MI5	MI6	MI7	MI8	MI9
VN	VN1	VN2	VN3	VN4	VN5	VN6	VN7	VN8	VN9
I1	I11	I12	I13	I14	I15	I16	I17	I18	I19
I2	I21	I22	I23	I24	I25	I26	I27	I28	I29
I3	I31	I32	I33	I34	I35	I36	I37	I38	I39
I4	I41	I42	I43	I44	I45	I46	I47	I48	I49
IV	IV1	IV2	IV3	IV4	IV5	IV6	IV7	IV8	IV9

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TABLE IX. Southern hemisphere – data series/zone codes

Data series	ZONE								
	A	B	C	D	E	F	G	H	J
GN	GNA	GNB	GNC	GND	N/A	N/A	GNG	N/A	GNJ
JN	JNA	JNB	JNC	JND	JNE	JNF	JNG	JNH	JNJ
ON	ONA	ONB	ONC	OND	ONE	ONF	ONG	ONH	ONJ
TP	TPA	TPB	TPC	TPD	TPE	TPF	TPG	TPH	TPJ
LF	LFA	LFB	LFC	LFD	LFE	LFF	LFG	LFH	LFJ
JG	JGA	JGB	JGC	JGD	JGE	JGF	JGG	JGH	JGJ
JA	JAA	JAB	JAC	JAD	JAE	JAF	JAG	JAH	JAJ
JR	JRA	JRB	JRC	JRD	JRE	JRF	JRG	JRH	JRJ
TF	TFA	TFB	TFC	TFD	TFE	TFF	TFG	TFH	TFJ
AT	ATA	ATB	ATC	ATD	ATE	ATF	ATG	ATH	ATJ
TC	TCA	TCB	TCC	TCD	TCE	TCF	TCG	TCH	TCJ
TL	TLA	TLB	TLC	TLD	TLE	TLF	TLG	TLH	TLJ
TT	TTA	TTB	TTC	TTD	TTE	TTF	TTG	TTH	TTJ
TQ	TQA	TQB	TQC	TQD	TQE	TQF	TQG	TQH	TQJ
CG	CGA	CGB	CGC	CGD	CGE	CGF	CGG	CGH	CGJ
CM	CMA	CMB	CMC	CMD	CME	CMF	CMG	CMH	CMJ
A1	A1A	A1B	A1C	A1D	A1E	A1F	A1G	A1H	A1J
A2	A2A	A2B	A2C	A2D	A2E	A2F	A2G	A2H	A2J
A3	A3A	A3B	A3C	A3D	A3E	A3F	A3G	A3H	A3J
MI	MIA	MIB	MIC	MID	MIE	MIF	MIG	MIH	MIJ
VN	VNA	VNB	VNC	VND	VNE	VNF	VNG	VNH	VNJ
I1	I1A	I1B	I1C	I1D	I1E	I1F	I1G	I1H	I1J
I2	I2A	I2B	I2C	I2D	I2E	I2F	I2G	I2H	I2J
I3	I3A	I3B	I3C	I3D	I3E	I3F	I3G	I3H	I3J
I4	I4A	I4B	I4C	I4D	I4E	I4F	I4G	I4H	I4J
IV	IVA	IVB	IVC	IVD	IVE	IVF	IVG	IVH	IVJ

The following are the valid <data series and zone> codes currently defined for RPF data that are zoned in the DTED zone format.

TABLE X. DTED zone format - data series/zone codes

Data Series	DTED Zone									
	1	2	3	4	5	6	7	8	9	A
D1[1]	D11	D12	D13	D14	D15	D16	D17	D18	D19	D1A
D2[1]	D21	D22	D23	D24	D25	D26	D27	D28	D29	D2A

[1] For user-specified applications of RPF that have not been approved for joint use

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5.1.6 Product data types

The following are the valid <product data type> values currently registered for RPF products.

TABLE XI. Product data types

Product Data Type	Product Name
ADRG	ARC Digitized Raster Graphics
ADRI	ARC Digital Raster Imagery
CADRG	Compressed ARC Digitized Raster Graphics
CIB	Controlled Image Base
CDTED[1]	Compressed Digital Terrain Elevation Data
DTED[1]	Digital Terrain Elevation Data

[1] For user-specified applications of RPF that have not been approved for joint use

5.1.7 Security country/international codes

For country codes that can be used to populate the 'security country/international code' field in the RPF Data Extension Segment (RPFDES), as well as the 'File Classification Authority' and 'File Release Instructions' fields in the other NITF 2.0 Headers and subheaders, see the 2-character codes found in the Geopolitical Entities and Codes (GEC) Register at https://nsgreg.nga.mil/genc/registers.jsp?register=GEC. Note, the GEC register is not to be confused with the GENC Register within the GENC standard.

5.1.8 Security classifications

The following <security classification> codes are defined for RPF.

TABLE XII. Classification codes

Code	Classification
T	Top Secret
S	Secret
C	Confidential
R	Restricted
U	Unclassified

5.1.9 Security release markings

The following <security release marking> codes are defined for RPF.

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TABLE XIII. Security release markings

Code	Release Marking
AT	ATOMAL (NATO designation to identify restricted or formerly restricted data)
BX	Background use only
PI	Caution - Proprietary information involved (PROPIN)
CC	Continued Control
PX	Copyright
CS	COSMIC (Caveat applied to NATO Top Secret)
CN	Critical Nuclear Weapon Design Information/DoD directive 5210.2 applies (CNWDI)
CR	Cryptographic (CRYPTO)
OR	Dissemination and Extraction Of information controlled by originator (ORCON)
DS	Distribution Limited-Destroy when no longer needed (LIMDIS)
TX	Encrypted for Transmission Only
FO	For Official Use Only (FOUO)
RF	Formerly restricted data (FRD)
IC	Intelligence Components Only
LA	Limited
LU	Limited official use
UB	NFIB (National Foreign Intelligence Board) departments only (NFIB ONLY)
DA	No Dissemination Abroad
NT	Noncompartmented
NC	Not releasable to contractors/consultants
FN	Not releasable to foreign national (DoD)
NF	Not releasable to foreign nationals (NOFORN)
NS	NSC Participating Agencies Only
IN	Personal Data - Privacy Act
EZ	Releasable to Republic of Vietnam
RD	Restricted Data (RD)
SA	SAO
SL	SAO Handling Caveat #1 (SAO-1)
HA	SAO Handling Caveat #2 (SAO-2)
HB	SAO Handling Caveat #3 (SAO-3)
SK	SAO Special Intelligence Handling Caveat #2 (SAO-SI-2)
HC	SAO Special Intelligence Handling Caveat #3 (SAO-SI-3)
HD	SAO Special Intelligence Handling Caveat #4 (SAO-SI-4)

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TABLE XIII. Security release markings - Continued

Code	Release Marking
SH	Single Integrated Operational Plan (SIOP)
SE	Single Integrated Operational Plan/Extremely Sensitive Information (SIOP/ESI)
SC	Special Control
SU	Special Handling Required/Not Releasable to foreign Nationals
SI	Special Intelligence (SI)
SN	Special Intelligence Handling Caveat #1 (SI-1)
EP	Authorized for release to Australia (REL AUSTRALIA)
EN	Authorized for release to Canada (REL CANADA)
ER	This information has been authorized for release to CENTO (REL CENTO)
BZ	Authorized for release to Chile (REL CHILE)
EK	Authorized for release to Denmark (REL DENMARK)
EQ	Authorized for release to Federal Republic of Germany (FEDERAL REPUBLIC OF GERMANY)
FR	Authorized for release to France (REL FRANCE)
IG	Authorized for release to Greece (REL GREECE)
FV	Authorized for release to Israel (REL ISRAEL)
FY	Authorized for release to Italy (REL ITALY)
PM	Authorized for release to Japan (REL JAPAN)
MR	Authorized for release to Jordan (REL JORDAN)
ET	Authorized for release to NATO (REL NATO)
FP	Authorized for release to NATO Forces (REL NATO FORCES)
FC	Authorized for release to New Zealand (REL NEW ZEALAND)
EF	Authorized for release to Norway (REL NORWAY)
FX	Authorized for release to Taiwan (REL TAIWAN)
EO	Authorized for release to Republic of Korea (REL REPUBLIC OF KOREA)
ES	This Information has been Authorized for release to SEATO (REL SEATO)
IS	Authorized for release to Denmark
FW	Authorized for release to Spain (REL SPAIN)
ED	Authorized for release to Sweden (REL SWEDEN)
PF	Authorized for release to Thailand (REL THAILAND)
EA	Authorized for release to Turkey (REL TURKEY)
WN	Authorized for release to United Kingdom (REL UNITED KINGDOM)
WI	Warning Notice - Intelligence sources or Methods Involved (WNINTEL)

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5.2 Table of contents data

This section defines codes that are used primarily in the RPF [table of contents file].

5.2.1 Producer codes

The following are the valid <producer code> values currently registered for RPF products.

TABLE XIV. Producer codes

ID	Producer code	Producer
1	AFACC	Air Force Air Combat Command
2	AFESC	Air Force Electronic Systems Center
3	ASG1	Allied System for Geospatial Intelligence (US)
4	ASG2	Allied System for Geospatial Intelligence (UK, AS-Melbourne)
5	ASG3	Allied System for Geospatial Intelligence (NZ, CA)
6	ASG4	Allied System for Geospatial Intelligence (AS-Bendigo)
7	SOCAF	Air Force Special Operations Command
8	SOCOM	United States Special Operations Command
9	PACAF	Pacific Air Forces
A	USAFE	United States Air Force, Europe
B	Non-DoD (NonDD)	US producer outside the Department of Defense
C	Non-US (NonUS)	All CADRG Produced for NATO by NATO member, includes overlays, annotations
D	NGA DCHUM (DCHUM)	NGA-produced overlay on NGA base frame
E	Non-NGA DCHUM (DCHMD)	Other US DoD-produced overlay on NGA base frame
F	Non-US DCHUM (DCHMF)	Non-US producer of overlay on ASG base frame
G	Non-DoD DCHUM (DCHMG)	US Non-DoD producer of overlay on NGA Base frame
H	IMG2RPF	Non-specified, Imagery formatted to RPF
I-Z		Reserved for future standardization

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5.2.2 Compression ratio code

The compression ratio code is an ASCII field within the [table of contents] file of RPF products. Examples of text that may be found in this field are given below.

TABLE XV. Compression ratio code

Code	Description
55:1	Digital product compressed approximately 55:1 with respect to the uncompressed data
12:1	Digital product compressed approximately 12:1 with respect to the uncompressed data
48:1	Digital product compressed approximately 48:1 with respect to the uncompressed data
UNC	Uncompressed RPF product
VAR	Variable compression ratio

5.3 Frame file data

This section defines codes that are used in the RPF [frame file].

5.3.1 Compression section

RPF data can be in a compressed or uncompressed format. This compression section defines valid codes for compression scheme parameters.

5.3.1.1 Compression algorithm ID

The following are the valid <compression algorithm id> values currently registered for RPF products.

TABLE XVI. Compression algorithm codes

ID	Compression Algorithm
1	VQ
2	JPEG Lossless[1]
3-65535	Reserved for future standardization

[1] For user-specified applications of RPF that have not been approved for joint use

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5.3.1.2 Compression lookup table ID

The following are the valid <compression lookup table id> values currently registered for RPF products.

TABLE XVII. Compression lookup table

ID	Compression Lookup Table
1	1x4 Row 0 VQ
2	1x4 Row 1 VQ
3	1x4 Row 2 VQ
4	1x4 Row 3 VQ
5	4x4 VQ
6	2x2 VQ
7	Huffman Code[1]
8	Length of Huffman Code[1]
9	Huffman Value[1]
10-65535	Reserved for future standardization

[1] For user-specified applications of RPF that have not been approved for joint use

5.3.1.3 Compression parameter ID

The <compression parameter id> is a code which identifies parameters which may be used in the [compression section] of an RPF [frame file]. Not all compression schemes require <compression parameter id> codes.

TABLE XVIII. Compression parameter codes

ID	Compression Parameter
1	JPEG data stream for Huffman table structure[1]
2-65535	Reserved for future standardization

[1] For user-specified applications of RPF that have not been approved for joint use

5.3.1.4 Compression parameter value

The <compression parameter value> can be either an explicit value to be used in a compression scheme or an index to a table used in the compression of RPF [frame file]s. Not all compression schemes require <compression parameter> values. Currently no <compression parameter> values are registered.

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5.3.2 Attribute section

The attribute section in RPF products defines ancillary or qualifying information about the data within the [frame file]. The following tables list currently registered attributes and parameters for RPF products.

TABLE XIX. Registered attributes and parameters

Attr ID	Attribute Description	Number of Parameters
1	Currency Date	1
2	Production Date	1
3	Significant Date	1
4	Map/Chart Source	4
5	Projection System	5
6	Vertical Datum	1
7	Horizontal Datum	1
8	Vertical Absolute Accuracy	2
9	Horizontal Absolute Accuracy	2
10	Vertical Relative Accuracy	2
11	Horizontal Relative Accuracy	2
12	Ellipsoid	1
13	Sounding Datum	1
14	Navigation System	1
15	Grid	1
16	Easterly Annual Magnetic Change	2
17	Westerly Annual Magnetic Change	2
18	Grid North-Magnetic North Angle	2
19	Grid Convergence Angle	2
20	Highest Known Elevation	4
21	Multiple Legend	1
22	Image Source Data	2
23[1]	Data Level	1
24	Chart Update Information	7
25	Contour Interval	2

[1] For user-specified applications of RPF that have not been approved for joint use

5.3.2.1 Attribute and parameter descriptions

The following are the valid parameters associated with attributes listed above and currently registered for RPF products.

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Currency Date (ID ::= 1)

Parameter ID	1	Description
Parameter Name	Currency Date	Date of most recent revision to the RPF product in the [frame file], in the form YYYYMMDD.
Data Type	asci:8	

Production Date (ID ::= 2)

Parameter ID	1	Description
Parameter Name	Production Date	Date that the source data was transformed to RPF format, in the form YYYYMMDD.
Data Type	asci:8	

Significant Date (ID ::= 3)

Parameter ID	1	Description
Parameter Name	Significant Date	Date that most accurately describes the basic date of the source product, in the form YYYYMMDD. It can be the completion date, compilation date, collection date, revision date, or other date depending on the product and circumstances.
Data Type	asci:8	

Map/Chart Source (ID ::= 4)

Parameter ID	1	Description
Parameter Name	Data series designation	A short title for the identification of a group of products usually having the same scale and/or cartographic specification (e.g., JOG 1501A, JOG 1501G, or JOG 1501C).
Data Type	asci:10	

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Parameter ID	2	Description The designation, within the data series, of the hardcopy source (e.g., if the hardcopy source is ONC G18, then the map designation is "G18".
Parameter Name	Map Designation	
Data Type	asci:8	
Parameter ID	3	Description Original horizontal datum of the hardcopy product. A list of horizontal datum codes is given in 5.3.2.2.
Parameter Name	Old Horizontal Datum Code	
Data Type	asci:4	
Parameter ID	4	Description Edition number of the source graphic, which uniquely identifies a product within a series and item number.
Parameter Name	Edition Identifier	
Data Type	asci:7	

Projection System (ID ::= 5)

Parameter ID	1	Description Projection of the source hardcopy graphic, as listed in 5.3.2.3.
Parameter Name	Projection Code	
Data Type	asci:2	
Parameter ID	2	Description The value of the first projection parameter. A definition of the first projection parameter is given in 5.3.2.3.
Parameter Name	Projection Parameter A	
Data Type	real:4	

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Parameter ID	3	Description The value of the second projection parameter. A definition of the second projection parameter is given in 5.3.2.3.
Parameter Name	Projection Parameter B	
Data Type	real:4	
Parameter ID	4	Description The value of the third projection parameter. A definition of the third projection parameter is given in 5.3.2.3.
Parameter Name	Projection Parameter C	
Data Type	real:4	
Parameter ID	5	Description The value of the fourth projection parameter. A definition of the fourth projection parameter is given in 5.3.2.3.
Parameter Name	Projection Parameter D	
Data Type	real:4	

[1] A full description of the projection systems and their parameters can be found in NGA.SIG.0012.

Vertical Datum (ID ::= 6)

Parameter ID	1	Description Vertical datum of this RPF [frame file] or areal extent, as listed in 5.3.2.2.
Parameter Name	Vertical Datum Code	
Data Type	asci:4	

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Horizontal Datum (ID ::= 7)

Parameter ID	1	Description
Parameter Name	Horizontal Datum Code	Horizontal datum of this RPF [frame file] or areal extent, as listed in 5.3.2.2.
Data Type	asci:4	

Vertical Absolute Accuracy (ID ::= 8)

Parameter ID	1	Description
Parameter Name	Vertical Absolute Accuracy	Vertical absolute accuracy of this RPF [frame file] or areal extent.
Data Type	uint:4	
Parameter ID	2	Description
Parameter Name	Accuracy Units of Measure	Units of measure for the vertical absolute accuracy, as listed in 5.3.2.6.
Data Type	uint:2	

Horizontal Absolute Accuracy (ID ::= 9)

Parameter ID	1	Description
Parameter Name	Horizontal Absolute Accuracy	Horizontal absolute accuracy of this RPF [frame file] or areal extent.
Data Type	uint:4	
Parameter ID	2	Description
Parameter Name	Accuracy Units of Measure	Units of measure for the horizontal absolute accuracy, as listed in 5.3.2.6.
Data Type	uint:2	

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Vertical Relative Accuracy (ID ::= 10)

Parameter ID	1	Description
Parameter Name	Vertical Relative Accuracy	Vertical relative accuracy of this RPF [frame file] or areal extent.
Data Type	uint:4	
Parameter ID	2	Description
Parameter Name	Accuracy Units of Measure	Units of measure for the vertical relative accuracy, as listed in 5.3.2.6.
Data Type	uint:2	

Horizontal Relative Accuracy (ID ::= 11)

Parameter ID	1	Description
Parameter Name	Horizontal Relative Accuracy	Horizontal relative accuracy of this RPF [frame file] or areal extent.
Data Type	uint:4	
Parameter ID	2	Description
Parameter Name	Accuracy Units of Measure	Units of measure for the horizontal relative accuracy, as listed in 5.3.2.6.
Data Type	uint:2	

Ellipsoid (ID ::= 12)

Parameter ID	1	Description
Parameter Name	Ellipsoid Code	Ellipsoid used in this RPF [frame file] or areal extent, as listed in 5.3.2.5.
Data Type	asci:3	

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Sounding Datum (ID ::= 13)

Parameter ID	1	Description
Parameter Name	Sounding Datum Code	Sounding datum used in this RPF [frame file] or areal extent, as listed in 5.3.2.2.
Data Type	asci:4	

Navigation System (ID ::= 14)

Parameter ID	1	Description
Parameter Name	Navigation System Code	Navigation system used in this RPF [frame file] or areal extent, as listed in 5.3.2.4.
Data Type	uint:2	

Grid (ID ::= 15)

Parameter ID	1	Description
Parameter Name	Grid Code	Grid code for this RPF [frame file] or areal extent, as listed in 5.3.2.7.
Data Type	asci:2	

Easterly Annual Magnetic Change (ID ::= 16)

Parameter ID	1	Description
Parameter Name	Easterly Annual Magnetic Change	Annual magnetic change in the easterly direction for this RPF [frame file] or areal extent.
Data Type	real:4	
Parameter ID	2	Description
Parameter Name	Units of Magnetic Change	Units for the magnetic change, as listed in 5.3.2.6.
Data Type	uint:2	

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Westerly Annual Magnetic Change (ID ::= 17)

Parameter ID	1	Description Annual magnetic change in the westerly direction for this RPF [frame file] or areal extent.
Parameter Name	Westerly Annual Magnetic Change	
Data Type	real:4	
Parameter ID	2	Description Units for the magnetic change, as listed in 5.3.2.6.
Parameter Name	Units of Magnetic Change	
Data Type	uint:2	

Grid North-Magnetic North Angle (ID ::= 18)

Parameter ID	1	Description Angle between north/south grid lines and magnetic north in this [frame file] or areal extent.
Parameter Name	Grid North-Magnetic North (G-M) Angle	
Data Type	real:4	
Parameter ID	2	Description Units of measure for the grid north-magnetic north angle, as listed in 5.3.2.6.
Parameter Name	Units of Angle	
Data Type	uint:2	

Grid Convergence Angle (ID ::= 19)

Parameter ID	1	Description Angle between grid north and true north in this [frame file] or areal extent.
Parameter Name	Grid Convergence Angle	
Data Type	real:4	

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Parameter ID	2	Description Units of measure for the grid convergence angle, as listed in 5.3.2.6.
Parameter Name	Units of Angle	
Data Type	uint:2	

Highest Known Elevation (ID ::= 20)

Parameter ID	1	Description Highest known elevation for the RPF frame or areal extent.
Parameter Name	Highest Known Elevation	
Data Type	real:8	
Parameter ID	2	Description Units of measure for the highest known elevation, as listed in 5.3.2.6.
Parameter Name	Units of elevation	
Data Type	uint:2	
Parameter ID	3	Description Latitude in decimal degrees of the highest known elevation point within this RPF [frame file] or areal extent.
Parameter Name	Latitude of elevation	
Data Type	real:8	
Parameter ID	4	Description Longitude in decimal degrees of the highest known elevation point within this RPF [frame file] or areal extent.
Parameter Name	Longitude of elevation	
Data Type	real:8	

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Multiple Legend (ID ::= 21)

Parameter ID	1	Description
Parameter Name	Legend File Name	Location of the legend files that apply to the source maps depicted in the file extents. Use directory name (e.g. map/chart series or city name + sheet number)
Data Type	asci:12	

Data Source (ID ::= 22)

Parameter ID	1	Description
Parameter Name	Data Source	Source of the data from which this RPF data was derived. For CIB: examples include "SPOT", "Landsat7". For CADRG: a map producer code, if available For the CDTED product, examples of this attribute include “imagery”, “carto”.1
Data Type	asci:12	
Parameter ID	2	Description
Parameter Name	GSD	For CIB: Ground Sample or post Distance of the RPF data. For CADRG: N/A
Data Type	uint:4	

Data Level (ID ::= 23)[1]

Parameter ID	1	Description
Parameter Name	Data Level	The level of the source for this data. For the CDTED product, this attribute would be used to distinguish between DTED/CDTED level 1 and DTED/CDTED level 2.
Data Type	uint:2	

[1] For-user specified applications of RPF that have not been approved for joint use

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Chart Update Information (ID : := 24)

Parameter ID	1	Description The total number of updates associated with this [frame file].
Parameter Name	Number of Updates	
Data Type	uint:2	
Parameter ID	2	Description The sequence number of this update.
Parameter Name	Update Number	
Data Type	Uint:2	
Parameter ID	3	Description The date associated with the update. This date is the date found in the OCF information, in the form YYYYMMDD.
Parameter Name	Update Date	
Data Type	asci:8	
Parameter ID	4	Description The number of subframes impacted by this update.
Parameter Name	Number of Subframes Impacted	
Data Type	uint: 2	
Parameter ID	5	Description An array of integers representing the list of subframes affected by the change. Each entry in the array is 2 bytes and the number of entries is specified by the “number of subframes”, indicated by parameter 4.
Parameter Name	List of Subframes	
Data Type	(uint:2) x (value of parameter 4)	

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Parameter ID	6	Description The number of characters in the description of the change.
Parameter Name	Number of Characters in Description	
Data Type	uint:2	
Parameter ID	7	Description A description of the change, as described in the OCF information. The length of this description is specified by the "Number of Characters in Description" indicated by parameter 6.
Parameter Name	Change Description	
Data Type	asci: var	

Contour Interval (ID ::= 25)

Parameter ID	1	Description Contour interval of the map or chart.
Parameter Name	Contour Interval	
Data Type	uint:2	
Parameter ID	2	Description Unit of measurement for the contour interval as specified in Section 5.3.2.6.
Parameter Name	Unit of Measurement	
Data Type	uint:2	

5.3.2.2 Datum codes

The following are the valid datum codes currently registered for RPF products, and as taken from NGA.STND.0036.

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TABLE XX. Horizontal datum codes

Code	Name (Region)
ADIM	Adindan (Mean value: Ethiopia and Sudan)
ADIA	Adindan (Ethiopia)
ADIB	Adindan (Sudan)
ADIC	Adindan (Mali)
ADID	Adindan (Senegal)
ADIF	Adindan (Cameroon)
ADIE	Adindan (Burkina Faso)
AFG	Afgooye (Somalia)
AIA	Antigua Island Astro 1943 Antigua (Leeward Islands)
AINA	Ain el Abd 1970 (Bahrain Island)
AINB	Ain el Abd 1970 (Saudi Arabia)
ANO	Anna 1 Astro (Cocos Islands)
ARFM	Arc 1950 (Mean value: Botswana, Lesotho, Malawi, Swaziland, Zaire, Zambia, and Zimbabwe)
ARFA	Arc 1950 (Botswana)
ARFB	Arc 1950 (Lesotho)
ARFC	Arc 1950 (Malawi)
ARFD	Arc 1950 (Swaziland)
ARFE	Arc 1950 (Zaire)
ARFF	Arc 1950 (Zambia)
ARFG	Arc 1950 (Zimbabwe)
ARFH	Arc 1950 (Burundi)
ARSM	Arc 1960 (Mean value: Kenya, Tanzania)
ARSA	Arc 1960 (Kenya)
ARSB	Arc 1960 (Tanzania)
ASC	Ascension Island 1958 (Ascension Island)
ATF	Astro Beacon "E" (Iwo Jima Island)
TRN	Astro Tern Is. 1961 (Tern Island, Hawaii)
SHB	Astro Dos 71/4 (St. Helena Island)
ASQ	Astro Station 1952 (Marcus Island)
AUA	Australian Geod. 1966 (Australia and Tasmania Is.)
AUG	Australian Geod. 1984 (Australia and Tasmania Is.)
PHA	Ayabelle Lighthouse (Djibouti)
IBE	Bellevue (IGN (Efate and Erromango Islands)
BER	Bermuda 1957 (Bermuda Islands)
BID	Bissau (Guinea-Bissau)
BOO	Bogota Observatory (Colombia)
BUR	Bukit Rimpah (Bangka and Belitum Islands, Indonesia)
CAZ	Camp Area Astro (Camp McMurdo Area, Antarctica)
CAI	Campo Inchauspe (Argentina)

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TABLE XX. Horizontal datum codes - Continued

Code	Name (Region)
CAO	Canton Astro 1966 (Phoenix Islands)
CAP	Cape (South Africa)
CAC	Cape Canaveral (Mean value: Florida and Bahama Islands)
CGE	Carthage (Tunisia)
CHI	Chatham 1971 (Chatham Island, New Zealand)
CHU	Chua Astro (Paraguay)
COA	Corrego Alegre (Brazil)
DAL	Dabola (Guinea)
BAT	Djakarta (Batavia) Sumatra Island, Indonesia)
GIZ	DOS 1968 0230 (Gizo Island, New Georgia Islands)
EAS	Easter Island 1967 (Easter Island)
EURM	European 1950 (Mean Value: Austria, Belgium, Denmark, Finland, France, Federal Republic of Germany, Gibraltar, Greece, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden and Switzerland)
EURA	European 1950 (Western Europe: Austria, Denmark, France, Federal Republic of Germany, Netherlands, and Switzerland)
EURB	European 1950 (Greece)
EURC	European 1950 (Norway and Finland)
EURD	European 1950 (Portugal and Spain)
EURE	European 1950 (Cyprus)
EURF	European 1950 (Egypt)
EURH	European 1950 (Iran)
EURI	European 1950 (Sardinia)
EURJ	European 1950 (Sicily)
EURK	European 1950 (England, Channel Islands, Ireland, Northern Ireland, Scotland, Shetland Islands, and Wales)
EURL	European 1950 (Malta)
EURS	European 1950 (Mean Value: Iraq, Israel, Jordan, Lebanon, Kuwait, Saudi Arabia, Syria)
EUS	European 1979 (Mean value: Austria, Finland, Netherlands, Norway, Spain, Sweden, and Switzerland)
FOT	Fort Thomas 1955 (Nevis, Saint Kitts, Leeward Islands)
GAA	GAN Datum (Addu Atoll, Republick of Maldives)

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TABLE XX. Horizontal datum codes - Continued

Code	Name (Region)
GEO	Geodetic Datum 1949 (New Zealand)
GUA	Guam 1963 (Guam Island)
GSE	Genung Segara (Kalimantan Island, Indonesia)
DOB	GUX 1 Astro (Guadalcanal Island)
HEN	Herat (North Afghanistan)
HJO	Hjorsey 1955 (Iceland)
HKD	Hong Kong 1963 (Hong Kong)
HTN	Hu-Tzu-Shan (Taiwan)
INFA	Indian 1954 (Thailand and Vietnam)
INHA	Indian 1975 (Thailand)
INDI	Indian (India and Nepal)
INDB	Indian (Bangladesh)
IRL	Ireland 1965 (Ireland and Northern Ireland)
ISG	ISTS 061 Astro 1968 (South Georgia Islands)
IST	ISTS 073 Astro 1969 (Diego Garcia)
JOH	Johnston Island 1961 (Johnston Island)
KAN	Kandawala (Sri Lanka)
KEG	Kerguelen Island 1949 (Kerguelen Island)
KEA	Kertau 1948 (West Malaysia and Singapore)
KUS	Kusaie Astro 1951 (Caroline Islands)
LCF	L.C. 5 Astro 1961 (Cayman Brac Island)
LEH	Leigon (Ghana)
LIB	Liberia 1964 (Liberia)
LUZA	Luzon (Philippines except Mindanao Island)
LUZB	Luzon (Mindanao Island, Philippines)
MPO	M'Poraloko (Gabon)
MIK	Mahe 1971 (Mahe Island)
MAR	Marco Astro (Salvage Islands)
MAS	Massawa (Eritrea, Ethiopia)
MER	Merchich (Morocco)
MID	Midway Astro 1961 (Midway Island)
MINA	Minna (Cameroon)
MIN	Minna (Nigeria)
ASM	Montserrat Island Astro 1958 (Montserrat, Leeward Islands)
NAHA	Nahrwan (Masirah Island, Oman)
NAHB	Nahrwan (United Arab Emirates)
NAHC	Nahrwan (Saudi Arabia)
NAP	Naparima (BWI Trinidad and Tobago)
NAS	North American 1927 (Mean Value)

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TABLE XX. Horizontal datum codes - Continued

Code	Name (Region)
NASC	North American 1927 (Mean value: CONUS)
NASA	North American 1927 (Eastern US)
NASB	North American 1927 (Western US)
NASD	North American 1927 (Alaska)
NASQ	North American 1927 (Bahamas, except San Salvador Island)
NASR	North American 1927 (San Salvador Island)
NASE	North American 1927 (Mean value: Canada)
NASF	North American 1927 (Alberta and British Columbia)
NASG	North American 1927 (Newfoundland, New Brunswick, Nova Scotia, and Quebec)
NASH	North American 1927 (Manitoba and Ontario)
NASI	North American 1927 (Northwest Territories and Saskatchewan)
NASJ	North American 1927 (Yukon)
NASO	North American 1927 (Canal Zone)
NASP	North American 1927 (Caribbean, Barbados, Caicos Islands, Cuba, Dominican Republic, Grand Cayman, Jamaica, Leeward Islands, and Turks Islands)
NASN	North American 1927 (Central America - Belize, Costa Rica, El Salvador, Guatemala, Honduras, and Nicaragua)
NAST	North American 1927 (Cuba)
NASU	North American 1927 (Hayes, Peninsula, Greenland)
NASL	North American 1927 (Mexico)
NAR	North American 1983 (Mean Value: Alaska, Canada, CONUS, Mexico, and Central America)
FLO	Observatorio 1966 (Corvo and Flores Islands, Azores)
OEG	Old Egyptian (Egypt)
OHAM	Old Hawaiian (Mean value)
OHAA	Old Hawaiian (Hawaii)
OHAB	Old Hawaiian (Kauai)
OHAC	Old Hawaiian (Maui)
OHAD	Old Hawaiian (Oahu)
FAH	Oman (Oman)
OGBM	Ord. Survey G. G. 1936 (Mean Value: England, Isle of Man, Scotland, Shetland, and Wales)
OGBA	Ord. Survey G.B. 1936 (England)
OGBB	Ord. Survey G.G. 1936 (England, Isle of Man, and Wales)

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TABLE XX. Horizontal datum codes - Continued

Code	Name (Region)
OGBC	Ord. Survey G.B. 1936 (Scotland and Shetland Islands)
OGBD	Ord. Survey G. B. 1936 (Wales)
PLN	Pico de las Nieves (Canary Islands)
PIT	Pitcairn Astro 1967 (Pitcairn Island)
PTB	Point 58 (Mean Value: Burkina Faso and Niger)
PON	Pointe Noire 1948 (Congo)
POS	Porto Santo 1936 (Porto Santo, Madeira Islands)
HIT	Prov. S. Chilean (S. Chile, 53 S.)
PRPM	Prov. S. Amer. 1956 (Mean value: Bolivia, Chile, Colombia, Ecuador, Guyana, Peru and Venezuela)
PRPB	Prov. S. Amer. 1956 (Northern Chile near 19 degrees south)
PRPC	Prov. S. Amer. 1956 (Southern Chile near 43 degrees south)
PRPD	Prov. S. Amer. 1956 (Columbia)
PRPE	Prov. S. Amer. 1956 (Ecuador)
PRPF	Prov. S. Amer. 1956 (Guyana)
PRPG	Prov. S. Amer. 1956 (Peru)
PRPH	Prov. S. Amer. 1956 (Venezuela)
PUR	Puerto Rico (Puerto Rico and Virgin Islands)
QAT	Qatar National (Qatar)
QUO	Qornog (South Greenland)
REU	Reunion (Mascarene Island)
MOD	Rome 1940 (Sardinia Island)
SAE	Santo (DOS) 1965 (Espirito Santo Island)
SAO	Sao Braz (Sao Miguel, Santa Maria Islands, Azores)
SAP	Sapper Hill 1943 (East Falkland Islands)
SCK	Schwarzeck (Namibia)
SGM	Selvagem Grande (Salvage Islands)
SANM	South American 1969 (Mean value: Argentina, Bolivia, Brazil, Chile, Columbia, Ecuador, Guyana, Paraguay, Peru, Trinidad and Tobago, and Venezuela)
SANA	South American 1969 (Argentina)
SANB	South American 1969 (Bolivia)
SANC	South American 1969 (Brazil)
SAND	South American 1969 (Chile)
SANE	South American 1969 (Columbia)
SANF	South American 1969 (Ecuador excluding Galapagos Islands)

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TABLE XX. Horizontal datum codes - Continued

Code	Name (Region)
SANG	South American 1969 (Guyana)
SANH	South American 1969 (Paraguay)
SANI	South American 1969 (Peru)
SANJ	South American 1969 (Ecuador including Baltra and Galapagos Islands)
SANK	South American 1969 (Trinidad and Tobago)
SANL	South American 1969 (Venezuela)
SOA	South Asia (Southeast Asia, Singapore)
POS	SE Base (Porte Santo) (Porto Santo & Madeira Islands)
GRA	SW Base (Faial, Graciosa, Pico, Sao Jorge, and Terceira Island, Azores)
TIL	Timbali 1948 (Brunei and East Malaysia - Sarawak and Sabah)
TAN	Tananarive Observatory 1925 (Madagascar)
TOK	Tokyo (Mean Value)
TOYM	Tokyo (Mean value: Japan, Korea, and Okinawa)
TOYA	Tokyo (Japan)
TOYB	Tokyo (Korea)
TOYC	Tokyo (Okinawa)
TDC	Tristan Astro 1968 (Tristan da Cunha)
UND	Undetermined (processed as if WGS 84)
MVS	Viti Levu 1916 (Viti Levu Island, Fiji Islands)
WAK	Wake Island Asto 1952 (Wake Atoll)
ENW	Wake-Eniwetok 1960 (Marshall Islands)
WGC	World Geodetic System 1972
WGE	World Geodetic System 1984
YAC	Yacare (Uruguay)
ZAN	Zanderij (Surinam)

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TABLE XXI. Sounding datum codes

Code	Description
MHW	Mean High Water
MHWN	Mean High Water Neaps
MHWS	Mean High Water Springs
MHHW	Mean Higher High Water
MLW	Mean Low Water
MLWN	Mean Low Water Neaps
MLWS	Mean Low Water Springs
MLLW	Mean Lower Low Water

TABLE XXII. Vertical datum codes

Code	Vertical Datum
MSL	Mean Sea Level (All elevations in the data set are referenced to the geoid of the specified datum.)
GEOD	Geodetic (All elevations in the data set are referenced to the ellipsoid of the specified datum.)

5.3.2.3 Projection codes and parameters

The following are the valid projection codes and parameters currently registered for RPF products.

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TABLE XXIII. Projection codes and parameters

Code Name	Parameters			
	A	**B**	**C**	**D**
AC Albers Equal Area	Central Meridian	Std. Parallel Nearest to Equator	Std. Parallel Farthest from Equator	Latitude of Origin
AL Azimuthal Equal Distant	Longitude of Tangency	Latitude of Tangency	---	---
RB Hotline Oblique Mercator	Longitude of Great Circle	Latitude of Great Circle	Azimuth of great Circle	---
LE Lambert Conformal Conic	Central Meridian	Std. Parallel Nearest to Equator	Std. Parallel Farthest from Equator	Latitude of Origin
MC Mercator	Central Meridian	Latitude of True scale	---	---
OC Oblique Mercator	Longitude of Great Circle	Latitude of Great Circle	Azimuth of Great Circle	---
OD Orthographic	Longitude of Tangency	Latitude of Tangency	---	---
PG Polar Stereographic	Central Meridian	Latitude of True Scale	Latitude of Origin	Longitude of Origin
PH Polyconic	Central Meridian	Latitude of Origin	---	---
TC Transverse Mercator	Central Meridian	Meridian of True North	Central Scale Factor	---
UT Universal Transverse Mercator	Central Meridian	---	---	---

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5.3.2.4 Navigation systems codes

The following are the valid navigation system codes currently registered for RPF products.

TABLE XXIV. Navigation system codes

Code	Description
0	Unknown
1	Circular Radio Beacon
2	CONSOL
3	DECCA
4	Radio Direction Finding
5	Direction Finding Beacon
6	Distance Finding
7	Long Range Air Navigation System (LORAN)
8	OMEGA
9	Undefined
10	Radar Response Beacon (RACON)
11	Radar
12	Radio
13	Radio Telephone
14	Undefined
15	TV
16	Microwave
17	Non-Directional Radio Beacon (NDB)
18	Non-Directional Beacon/Distance Measuring Equipment (NDB/DME)
19	Radio Range (RNG)
20	VHF Omni Directional Radio Range (VOR)
21	VHF Omni Directional (VOR/DME)
22	VHF Omni Directional (VORTAC)
23	Tactical Air Navigation Equipment (TACAN)
24	Instrument Landing System (ILS)
25	(ILS/DME)
26	Localizer (LOC)
27	Localizer/Distance measuring Equipment (LOC/DME)
28	Simplified Direction Facility
29	Landing Distance Available (LDA)
30	Microwave Landing System (MLS)
31	Fan Marker
32	Bone Marker

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TABLE XXIV. Navigation system codes - Continued

Code	Description
33	Radio Telegraph
34	Ground Control Approach (GCA)
35	Radar Antenna
36	Undefined
37	Precision Approach Radar (PAR)
38	Aeronautical Radio
39	Undefined
40	Radio Beacon
41	Rotating Loop Radio Beacon
42	Visual Flight Rules (VFR) Test Signal Maker
43	Undefined
44	Consol Radio Beacon
45	Aeronautical Radio Beacon
46	Radar Station
47	Hifix
48	Hyperfix
999	Other

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5.3.2.5 Ellipsoid codes

The following are the valid ellipsoid codes currently registered for RPF products.

TABLE XXV. Ellipsoid codes

Code	Description
AAY	Airy 1830
AUN	Australian National
BES	Bessel 1841
bes	Bessel 1841 (Namibia)
CLK	Clark 1866
CLJ	Clark 1880
EVE	Everest
FM	Fischer 1960
FC	Fischer 1968
RE	Geodetic Reference System 1967
GRS	Geodetic Reference System 1980
HE	Helmert 1906
HO	Hough 1960
INT	International 1924
KRA	Krasovsky
AAM	Modified Airy
EVM	Modified Everest
FA	Modified Fischer 1960 (South Asia)
SA	South American 1969
WGA	World Geodetic System 1960
WGB	World Geodetic System 1966
WGC	World Geodetic System 1972
WGD	World Geodetic System 1984

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5.3.2.6 Units of measure codes

The following are the valid units of measure codes currently registered for RPF products.

TABLE XXVI. Unit of measure codes

Code	Description
0	Unknown
1	Meters (Length, Area, Volume)
2	Kilopounds (Weight)
3	Seconds (Of Time)
4	Meters/Second (Speed)
5	Cubic Meters/Second (Flow)
6	Volt (Electrical Potential)
7	Watt (Ten to the Sixth (Electrical Power))
8	Hertz (Frequency)
9	± DDD MM SS.S (Angular)
10	± HH MM SS.S (Time)
11	International Nautical Mile (1852 meters)
12	Knot
13	Nautical Mile/Day
14	Feet
15	Fathom
16	Micrometers (Microns)
17	Mils
18	Seconds of arc
19	Minutes of arc
20	Degrees of arc
999	Other

5.3.2.7 Grid codes

The following are the valid grid codes currently registered for RPF products.

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TABLE XXVII. Grid codes

Code	Description
AD	Aden Zone
AF	Afghanistan Gauss-Krueger Grid
AG	Air Defense Grid
AI	Air Support Grid
AJ	Alabama Coordinate System
AK	Alaska Coordinate System
AL	Algeria Zone
AM	Albania Bonne Grid
AN	Alpha-Numeric (Atlas) Grid
AO	Arbitrary Grid
AP	American Samoa Coordinate System
AQ	Argentina Gauss-Krueger Conformal Grid
AR	Artillery Referencing System
AS	Arizona Coordinate System
AU	Australia Belts
AV	Arkansas Coordinate System
AW	Australian Map Grid
AX	Azores Gauss Conformal Grid
AZ	Azores Zone
BA	Baku 1927 Coordinate System
BB	Bavaria Solder Coordinate System
BC	Belgium-Lambert Grid
BE	Belgium Bonne Grid
BF	Brazil Gauss Conformal GridBO
BO	Borneo Rectified Skew Orthomorphic Grid
BW	British West Indies Grid
CB	California Coordinate System
CD	Canada British Modified Grid
CE	Ceylon Belt (Transverse Mercator)
CF	Canary Islands (Spanish) Lambert Grid
CG	Chile Gauss Conformal Grid
CH	China Belt
CI	Canary Islands Zone
CJ	China Lambert Zone
CK	Colorado Coordinate System
CM	Connecticut Coordinate System
CN	Caspian Zone
CO	Costa Rica Lambert Grid
CQ	Crimea Grid
CR	Crete Zone

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TABLE XXVII. Grid codes - Continued

Code	Description
CT	Cuba Lambert Grid
CU	Caucasus Zone
CV	Cape Verde Islands Zone
CW	British Cassini Grid
CX	Czechoslovak Uniform Cadastral Coordinate System
CY	Cyprus Grid
CZ	Czechoslovak Military Grid
DA	Danube Zone
DB	Dahomey Belt
DC	Denmark General Staff Grid
DD	Delaware Coordinate System
DE	Dominican Lambert Grid
DJ	Denmark Geodetic Institute System 1934
DK	Cape Verde Peninsula Grid
EA	East Africa Belt
EB	English Belt
ED	Egypt Gauss Conformal Grid
EE	El Salvador Lambert Grid
EF	Estonian Grid
EP	Egypt Purple Belt
ER	Egypt Red Belt
ET	Egypt 35 Degree Belt
FA	Fernando Poo Gauss Grid
FB	Fiji Grid
FC	Florida Coordinate System
FD	French Bonne Grid
FE	French Guiana Gauss Grid
FF	French Somaliland Gauss-LaBorde Grid
FI	French Indochina Grid
FJ	Franz Josef Land Zone
FL	French Lambert Grid
FO	Formosa (Taiwan) Gauss-Schreiber Coordinate System
FS	French Equatorial Africa Grid
GA	Gabon Belt
GB	Italy Gauss-Boaga Grid (Transverse Mercator)
GD	Gabon Gauss Conformal Grid
GE	World Geographic Reference System (GEOREF)

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TABLE XXVII. Grid codes - Continued

Code	Description
GF	Guadeloupe Gauss-LaBorde Grid
GG	Colombia Gause Conformal Grid
GH	Sweden Gauss-Hanover Grid
GI	Georgia Coordinate System
GK	Gauss-Krueger Grid (Transverse Mercator)
GL	Greece Azimuthal Grid
GN	German Army Grid (DHG)
GO	Ghana National Grid
GP	Greece Bonne Grid
GQ	Greece Conical Mecklenburg Coordinates
GR	Greece Conical Mecklenburg Coordinate (New Numbering)
GT	Greenland Lambert Grid
GU	Guinea Zone
GV	Guam Coordinate System
GW	Guatemala Lambert Grid
GY	Guyana Transverse Mercator Grid
HB	Haiti Lambert Grid
HC	Hawaii Coordinate System
HD	Hawaii Grid
HE	Honduras Lambert Grid
HF	Hong Kong New System Cassini Grid
HG	Hungary Stereographic Grid
HK	Hong Kong Colony Grid
IA	Idaho Coordinate System
IB	Illinois Coordinate System
IC	Indiana Coordinate System
ID	Indonesia Mercator Grid
IE	Indonesia Polyhedric Grid
IF	Iowa Coordinate System
IG	Ivory Coast Azimuthal Grid
IH	Irish Cassini Grid
IJ	Ivory Coast Belt
IK	Irish Transverse Mercator Grid
IL	Iceland New Lambert Zone
IN	India Zone
IR	Iraq National Grid
IT	Italy Zone
IY	Ivy - Found on an HA in Marshall Islands
IZ	Iceland Zone

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TABLE XXVII. Grid codes - Continued

Code	Description
JA	Jamaica Foot Grid
JB	Japan Plane-Rectangular Coordinate System
JC	Japan Gauss-Schreiber Grid
JO	Johore Grid
KA	Austria Gauss-Krueger Grid
KB	Bulgaria Gauss-Krueger Grid
KC	Katanga Grid
KD	Kansas Coordinate System
KE	Kentucky Coordinate System
KF	Finland Gauss-Kreuger Grid
KG	German Gauss-Kreuger Grid
KH	Kenya Colony Grid
KJ	Korea Gauss-Schreiber Coordinate System
KK	Louisiana Coordinate System
KL	Lithuania Gauss-Kreuger Grid
KN	Kwantung Province Grid
KT	Turkey Gauss-Kreuger Grid
KW	Kwangsi Province Grid
KX	Luxembourg Gauss-Kreuger Grid
LC	Lambert Conformal Conic Grid
LD	Latvia Coordinate System
LE	Levant Zone
LF	Levant Stereographic Grid
LG	Liberia Rectified Skew Orthomorphic Grid
LI	Libya Zone
LL	Sirte (Libya) Lambert Grid
MA	Malaya Grid
MB	Malta Belt
MC	Maldive-Chagos Belt
MD	Madiera Zone
ME	Mediterranean Zone
MF	Maine Coordinate System
MG	Malaya Rectified Skew Orthomorphic (Yard) Grid
MH	Martinique Gauss Grid
MI	Maryland Coordinate System
MJ	Massachusetts Coordinate System
MK	Mexican Lambert Grid
ML	Michigan Coordinate System
MM	Mecca-Muscat Zone
MN	Minnesota Coordinate System

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TABLE XXVII. Grid codes - Continued

Code	Description
MO	Madagascar Grid (Laborde)
MP	Mississippi Coordinate System
MQ	Morocco Zone
MT	Missouri Coordinate System
MU	Mauritius Zone
MV	Montana Coordinate System
MW	Mozambique Lambert Grid
MX	Mozambique Polyconic Grid
NA	Northwest Africa Zone
NC	Nigeria Colony Belt
ND	National Grid of Great Britain
NE	Northern European Zone
NF	Nebraska Coordinate System
NG	Numeric Grid
NI	Niger Zone
NJ	Netherlands Stereographic Grid (Old Numbering)
NK	North Korea Gauss-Kreuger Grid
NL	Netherlands Stereographic Grid (New Numbering)
NM	Netherlands East Indies Equatorial Zone British Metric Grid (Lambert)
NO	Nord de Guerre Zone
NP	Nevada Coordinate System
NQ	New Sierra Leone Colony Grid
NR	New York Coordinate System
NS	Netherlands East Indies Southern Zone
NT	New Zealand National Grid
NU	Nicaragua Lambert Grid
NV	Niger Belt
NW	North Caroline Coordinate System
NX	North Dakota Coordinate System
NY	Netherlands East Indies Equatorial Zone U.S. Yard Grid
NZ	New Zealand Belt
OA	Northern Malaya Grid
OB	Norway Gauss-Kreuger Grid
OD	Ohio Coordinate System
OE	Oklahoma Coordinate System
OR	Orange Report Net
OS	Oregon Coordinate System
PA	Palestine Belt

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TABLE XXVII. Grid codes - Continued

Code	Description
PB	Panama Lambert Grid
PC	Palestine Civil Grid (Cassini)
PD	Paraguay Gauss-Kreuger Grid
PE	Peiping Coordinate System of 1954
PF	Pennsylvania Coordinate System
PI	Peru Polyconic Grid
PJ	Philippine Plane Coordinate System
PK	Poland Gauss-Kreuger Grid
PL	Poland Quasi-Stereographic Grid
PP	Philippine Polyconic Grid
PQ	Portugal Bonne Grid, Old
PR	Portugal Bonne Grid, New
PS	Portugal Gauss Grid
PT	Puerto Rico Coordinate System
PU	Puerto Rico Lambert Grid
QA	Qatar Grid
QU	Qatar Peninsula Grid
RB	Russian Belt
RC	Reunion Gauss Grid
RD	Rhode Island Coordinate System
RE	Rumania Bonne Grid
RF	Soviet Coordinate System of 1942
RH	Rumania Lambert-Cholesky Grid
RI	Rumania Stereographic Grid
RT	Pulkovo Coordinate System of 1932
SA	South Africa Belt
SB	Senegal Gauss Conformal Grid (Belt)
SD	South Africa Coordinate System (Republic of South Africa)
SE	Senegal Belt
SF	South Carolina Coordinate System
SH	Sahara Zone
SI	South Dakota Coordinate System
SJ	South Libya Zone
SK	Sarawak Grid
SL	Spain Lambert Grid
SN	Southern New Guinea Grid
SQ	South Georgia Lambert Grid
SR	South Syria Lambert Grid
SS	Spanish North-Morocco Lambert Grid

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TABLE XXVII. Grid codes - Continued

Code	Description
SV	Svalbard Gauss-Kreuger Grid
SX	Svobodny 1935 Coordinate System
SY	Seychelles Belt
SZ	Spitzbergen Zone
TA	Tanganyika Territorial Grid
TB	Tashkent 1875 Coordinate System
TC	Tennessee Coordinate System
TD	Texas Coordinate System
TE	Tobago Grid
TF	Trinidad Grid
TG	Trucial Coast Cassini Grid
TH	Trucial Coast Transverse Mercator Grid
TI	Turkey Bonne Grid
TN	Tunisia Zone
UA	Uganda Cassini Coordinate System
UB	Unidentified Grid
UC	Uruguay Gauss-Kreuger Grid
UD	Utah Coordinate System
UP	Universal Polar Stereographic System
US	U.S. Polyconic Grid System
UT	Universal Transverse Mercator Grid
VA	Vermont Coordinate System
VB	Virginia Coordinate System
VE	Venezuela Modified Lambert Grid
VI	Viet Nam Azimuthal Grid
WA	West Malaysia Rectified Skew Orthomorphic (Metric) Grid
WB	Switzerland Bonne Grid
WC	Switzerland Conformal Oblique Cylindrical Grid
WD	West Virginia Coordinate System
WE	Wisconsin Coordinate System
WF	Wyoming Coordinate System
WP	World Polyconic System
YA	Yugoslavia Gauss-Kreuger Grid (Not Reduced)
YU	Yunnan Province Grid

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5.3.3 Relationship code

The <relationship> code describes the relationship between two or more RPF [frame file]s. Valid codes are listed below.

TABLE XXVIII. Relationship codes

Code	Description
1 2	Stereographic pairs Legend Image
3-65535	Reserved for future standardization

5.4 NITF Registered Data Tags

The following codes are valid NITF registered data tags for use by RPF products that are written in an NITF message. The official register of tagged data extensions is located under the RPF group at http://jitc.fhu.disa.mil/projects/nitf/registers/trereg.aspx.

Changes to this group are managed through the same DoD and NGA processes as the codes listed in this document.

TABLE XXIX. NITF registered data tags

Tag	Description
RPFHDR	User Defined Header Section for RPF
RPFIMG	User Defined Image Section for RPF
RPFDES	Data Extension Segment for RPF

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use

This document is intended for use with MIL-STD-2411, Military Standard for Raster Product Format (RPF), and all specifications governed by that military standard. Identification codes and values tabulated in this document are the registered data values to be used in all products conforming to the RPF standard. The registered data values are included in this document, separate from MIL-STD-2411 so that new values can easily be added to the registered data set without modifying the governing standard.

6.2 Acquisition requirements

When this standard is used in acquisition, the applicable issue of DODISS must be cited in the solicitation (see 2.1.1 and 2.2).

6.3 Subject term (key word) listing

Scanned Map

6.4 Changes from previous issue

See the summary of change 4 modifications on page iii. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
NGA - MP

Preparing activity:
NGA - MP
(Project GINT-2020-002)

Review activities:
Air Force - 09
Navy - NO
Marine Corps - MC
DISA - DC2

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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